



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2020-2021
HALF YEARLY EXAMINATION (ONLINE)

CLASS: IX
SUBJECT: COMPUTER APPLICATIONS

FULL MARKS: 60
TIME: 1 $\frac{1}{2}$ HOURS

Instructions:

- All questions are compulsory.
- This paper consists of three printed pages.
- All working including rough work must be clearly shown on the same sheet as the rest of the answers.

Section A (15 Marks)
[Attempt all questions.]

Question 1

[5x1=5]

- a) Discuss the working of a shorthand operator with the help of an example.
- b) Which data type(s) is an exact representation of fractional values? What is the memory space in bits occupied by the same?
- c) What is a pure expression? Explain with the help of an example.
- d) Write a Java statement to display the following sentence

“Hello” the path of success is drive:\ hard work\success

- e) Rewrite the following program segment using ternary operator:

```
if (a + b < c || a + c <= b || b + c <= a)
    s = "Triangle is not possible";
else
    s = "Triangle is possible";
```

Question 2

[5x2=10]

- a) Predict the output of the following code, where, x= 1.5, y= -3.0;
 - i) `System.out.println(Math.floor(Math.round(y))+Math.min(x,y));`
 - ii) `System.out.println(Math.ceil(Math rint(x)/Math.max(y,x)));`

- b) Write the equivalent java expression for the following: $\frac{1}{2x^2i} (x - i^2) + ky^3$

c) What will be the output of the following code:

```
int k = -10; float j = 1.3f;
k = ++k * --j + k / j - --k % j++ ;
System.out.println(k);
System.out.println(j);
```

d) State the final value of p at the end of the following program segment. How many times the loop will execute?

```
int p = 4;
while (p<=15)
{ p++;
  if(p== 10)
    continue;
  System.out.println(p++); }
```

e) Rewrite the following program using while:

```
class Test
{ public static void main( )
  { int x,c;
    for(x=10,c=20;c>=10;c=c-2)
    { x++;
      System.out.println(x); }}
```

Section B (45 Marks)

[Attempt all questions.]

The answers in this Section should consist of the Programs in BlueJ environment . Each program should be written using Variable descriptions/Mnemonic Codes so that the logic of the program is clearly depicted. Flow-Charts and Algorithms are not required.]

Question 3

[15]

Write a program to input a set of numbers (including positive and negative).

Perform the following tasks:

- (a) Count the positive numbers
- (b) Count the negative numbers
- (c) Sum of positive numbers
- (d) Sum of negative numbers

The program should terminate when the user enters digit 0.

Question 4**[15]**

Write a menu driven program to calculate and display the sum of the following series :

a) $S = 2 - 4 + 6 - 8 + \dots - 20$

b) $S = (1 \sqrt{3}/a^2) + (3 \sqrt{5}/a^3) + (5 \sqrt{7}/a^4) + \dots \text{ to } n$

Question 5**[15]**

An Electricity Board charges for electricity per month from their consumers according to the units consumed. The tariff is given below:

Units Consumed	Charges
Up to 200 units	₹3.80/unit
More than 200 units and up to 300 units	₹4.40/unit
More than 300 units and up to 400 units	₹5.10/unit
More than 400 units	₹5.80/unit

Write a program to calculate the electricity bill by taking consumer's name and units consumed as inputs. Display the output in a tabular format.

Consumer name	Units consumed	Total bill
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